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SCENARIO-BASED ASSIGNMENT

DEEP LEARNING-ASSESSMENT-2

1. Machine Learning vs Deep Learning

A healthcare company wants to develop an AI system to detect pneumonia from chest X-ray images. Initially, the development team plans to use traditional Machine Learning with manually extracted image features. However, after collecting millions of medical images, they decide to switch to a Deep Learning approach.

Question: Compare Machine Learning and Deep Learning in the context of this scenario. Justify why Deep Learning is a better choice for this application by discussing data requirements, feature extraction, computational resources, accuracy, and real-world implementation challenges.

Machine Learning (ML) and Deep Learning (DL) are both Artificial Intelligence techniques, but they differ in how they learn from data. In this healthcare scenario, ML initially uses manually extracted features from chest X-ray images, such as lung texture and shape, to detect pneumonia. However, after collecting millions of medical images, Deep Learning becomes a better choice because it can automatically learn features directly from raw images.

Deep Learning requires a large amount of data, whereas Machine Learning performs well even with smaller datasets. Since the company has millions of X-ray images, Deep Learning can learn more complex patterns and improve its performance. Unlike ML, which depends on human experts for feature extraction, Deep Learning automatically identifies important features through multiple neural network layers.

Deep Learning also requires powerful computational resources such as GPUs or TPUs, while Machine Learning can often be trained using CPUs. Although Deep Learning takes more time and computational power to train, it generally achieves much higher accuracy in image classification tasks such as medical diagnosis.

In real-world implementation, Machine Learning faces challenges like manual feature engineering and limited scalability, whereas Deep Learning requires large labeled datasets, expensive hardware, and longer training time. Despite these challenges, Deep Learning is the preferred approach because it provides better accuracy, automatically extracts features, and performs exceptionally well on large-scale medical image datasets.

2. Representation Learning

A retail company has millions of customer purchase records and wants to automatically identify customer buying patterns without manually selecting important features. The company plans to use a Deep Learning model that can learn meaningful representations directly from raw transactional data.

Question: Explain the concept of Representation Learning and discuss how it helps solve the company's problem. Compare it with traditional feature engineering and explain its advantages in handling complex realworld datasets.

Representation Learning is a Deep Learning technique in which the model automatically learns useful features from raw data without requiring manual feature engineering. In the given retail scenario, the company has millions of customer purchase records and wants to identify buying patterns automatically. Representation Learning enables the neural network to discover meaningful patterns directly from the transactional data.

Traditional Machine Learning requires experts to manually select important features such as customer age, purchase frequency, spending amount, and product categories. This process is time-consuming and may overlook hidden relationships in the data.

In contrast, Representation Learning automatically identifies complex customer behaviors, purchasing habits, seasonal trends, and product relationships. It learns multiple levels of data representations, improving the model's ability to make accurate predictions.

The major advantages of Representation Learning include automatic feature extraction, better handling of large and complex datasets, improved prediction accuracy, reduced dependence on domain expertise, and the ability to capture nonlinear relationships among data. Therefore, Representation Learning is an effective solution for analyzing large real-world datasets and discovering hidden customer buying patterns.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system. Two engineers propose different neural network architectures. One suggests using a very wide network with a single hidden layer containing thousands of neurons, while the other recommends a deeper network with multiple hidden layers and fewer neurons per layer.

Question: Analyze both approaches by explaining the concepts of width and depth in neural networks. Discuss how each architecture affects feature learning, computational complexity, model performance, and generalization. Recommend the most suitable architecture with proper justification.

Width refers to the number of neurons in a hidden layer, while depth refers to the number of hidden layers in a neural network.

A very wide neural network with a single hidden layer contains many neurons and can learn several features simultaneously. It is relatively simple to design but requires a large number of parameters, consumes more memory, and is more likely to overfit the training data. It also has limited ability to learn hierarchical features.

A deep neural network consists of multiple hidden layers with fewer neurons in each layer. The first layers learn simple features such as edges, the middle layers learn shapes and patterns, and the deeper layers learn high-level features such as handwritten digits. This hierarchical learning improves feature extraction and overall model performance.

Although deep networks require more computational resources and longer training time, they generally provide better accuracy and better generalization on unseen data.

For handwritten digit recognition, a deep neural network is the most suitable architecture because it learns complex hierarchical features efficiently, reduces overfitting, and achieves higher recognition accuracy compared to a very wide shallow network.

4. Activation Functions

A company is building a Deep Learning model for image classification. During training, the team observes problems such as slow convergence, vanishing gradients, dead neurons, and unstable outputs in the final classification layer.

Question: Evaluate the suitability of Sigmoid, Tanh, ReLU, Leaky ReLU, ELU, and Softmax activation functions for different layers of the neural network. Explain which activation function should be used in each stage of the model and justify your choices based on their characteristics.

Activation functions introduce non-linearity into neural networks and help them learn complex patterns.

Sigmoid produces outputs between 0 and 1 and is mainly used in the output layer for binary classification. However, it suffers from the vanishing gradient problem and slows down learning.

Tanh produces outputs between -1 and 1, making it zero-centered and generally better than Sigmoid for hidden layers. However, it can still experience vanishing gradients.

ReLU (Rectified Linear Unit) outputs zero for negative inputs and the input value for positive inputs. It trains faster, reduces the vanishing gradient problem, and is the most commonly used activation function in hidden layers. However, it may suffer from the dead neuron problem.

Leaky ReLU solves the dead neuron problem by allowing a small gradient for negative inputs, making learning more stable.

ELU (Exponential Linear Unit) improves learning by producing smooth negative outputs, reducing bias shift, and enabling faster convergence in deep networks.

Softmax converts output values into probability distributions that sum to one and is used in the output layer for multi-class classification problems.

For the given image classification model, ReLU should be used in the hidden layers, Leaky ReLU or ELU can be used if dead neurons or unstable learning occur, and Softmax should be used in the final output layer for multi-class image classification.

5. Unsupervised Training of Neural Networks

A social media platform collects billions of unlabeled user images every month. Since manually labeling such a massive dataset is expensive and time-consuming, the company wants to train a neural network that can automatically discover hidden patterns and useful representations.

Question: Explain how unsupervised training of neural networks can address this problem. Discuss the learning process, advantages over supervised learning in this scenario, and describe suitable techniques that can be applied for representation learning.

Unsupervised training is a learning approach in which neural networks are trained using unlabeled data. Since the social media platform collects billions of unlabeled images, manually assigning labels is expensive and impractical. Unsupervised learning enables the network to automatically discover hidden patterns and useful representations from the available data.

During training, the model analyzes similarities and differences among images and learns meaningful internal representations without requiring class labels. These learned representations can later be used for image search, recommendation systems, clustering, and image classification.

Compared to supervised learning, unsupervised learning eliminates the need for manual labeling, reduces costs, utilizes massive amounts of available data, and learns more general and transferable features.

Suitable techniques for representation learning include **Autoencoders**, which learn compressed representations of images; **Variational Autoencoders (VAEs)**, which learn probabilistic latent features; **Self-Supervised Learning**, which creates learning tasks from the data itself; **Contrastive Learning**, which learns by comparing similar and different images; and **Clustering methods**, which group similar images based on learned features.

Therefore, unsupervised training is an effective solution for learning meaningful image representations from large-scale unlabeled datasets while reducing annotation costs and improving performance in real-world applications.